

## Experiment No. 08

### DC to DC Converter

**Aim:** To study the operation of DC - DC converters using MATLAB Simulink.

Converter topologies:

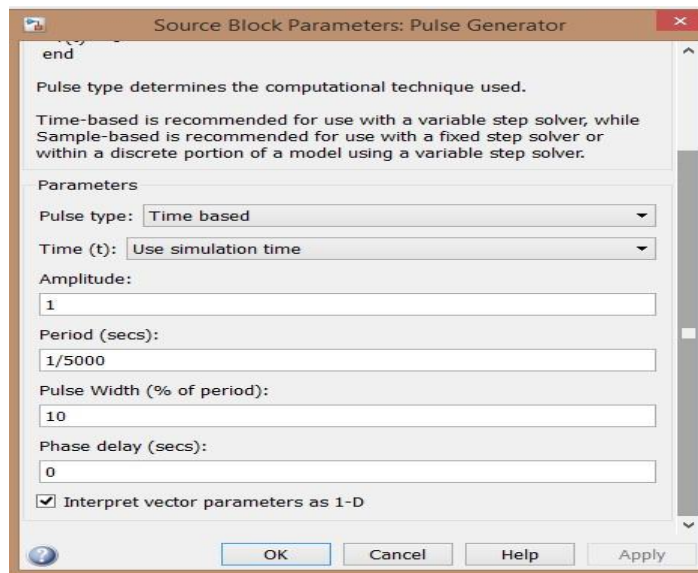
- i) Buck Converter,
- ii) Boost Converter and
- iii) Buck-Boost Converter

#### **Simulation tools required:**

MATLAB 2015a

#### **Simulation Procedure:**

1. Construct the circuit in the MATLAB, Simulink,
2. Use pulse generator to generate the gate pulses to the IGBT of the DC-DC converter,
3. Configure the pulse generator as shown below



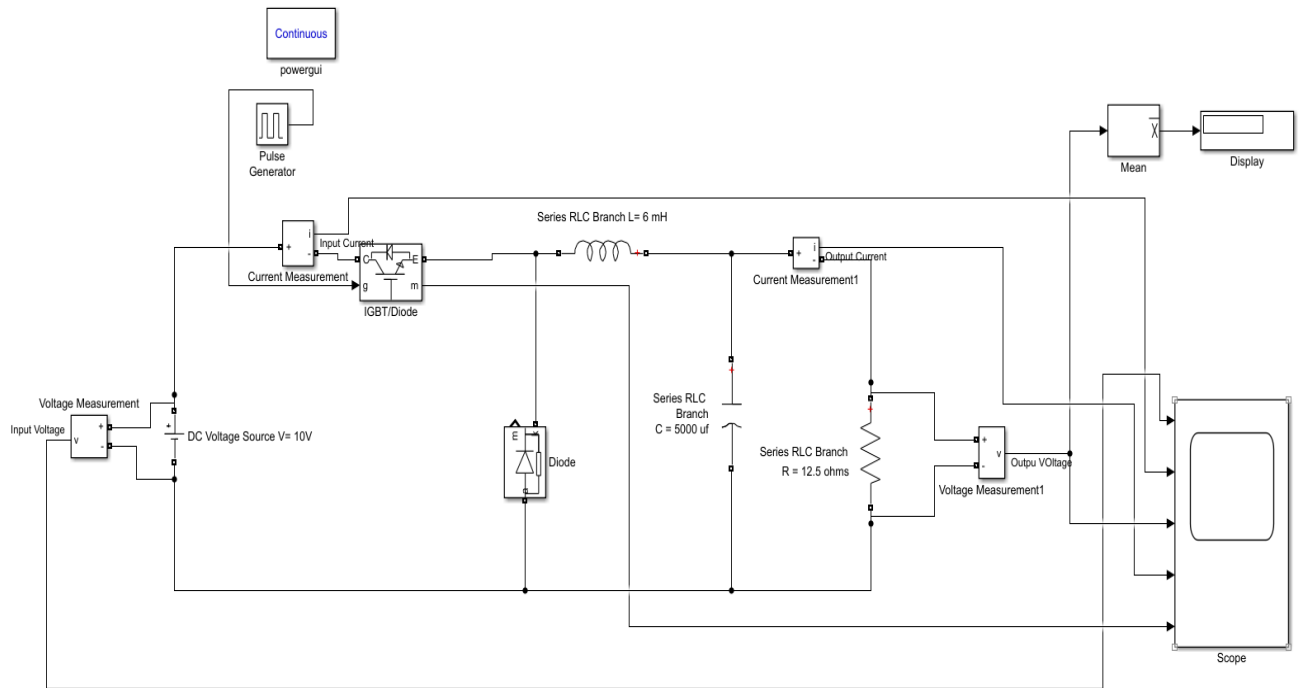
4. The duty cycle of the converter is varied by changing the Pulse width in the pulse generator.
5. Vary the duty cycle as shown in observation table above and note the output voltage values.
6. Use the output voltage formula of each converter type to compute the output voltage calculated in the observation table. For example, considering a

1. Buck-Converter, the output voltage can be expressed as  $V_o = D V_{in}$ ,

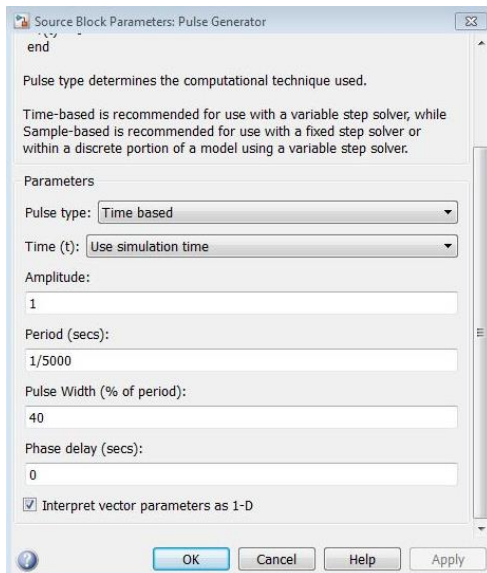
2. Boost-Converter, the output voltage can be expressed as  $V_o = \frac{1}{(1-D)} (V_{in})$

3. Buck-Boost Converter, the output voltage can be expressed as  $V_o = \frac{D}{D-1} V_{in}$

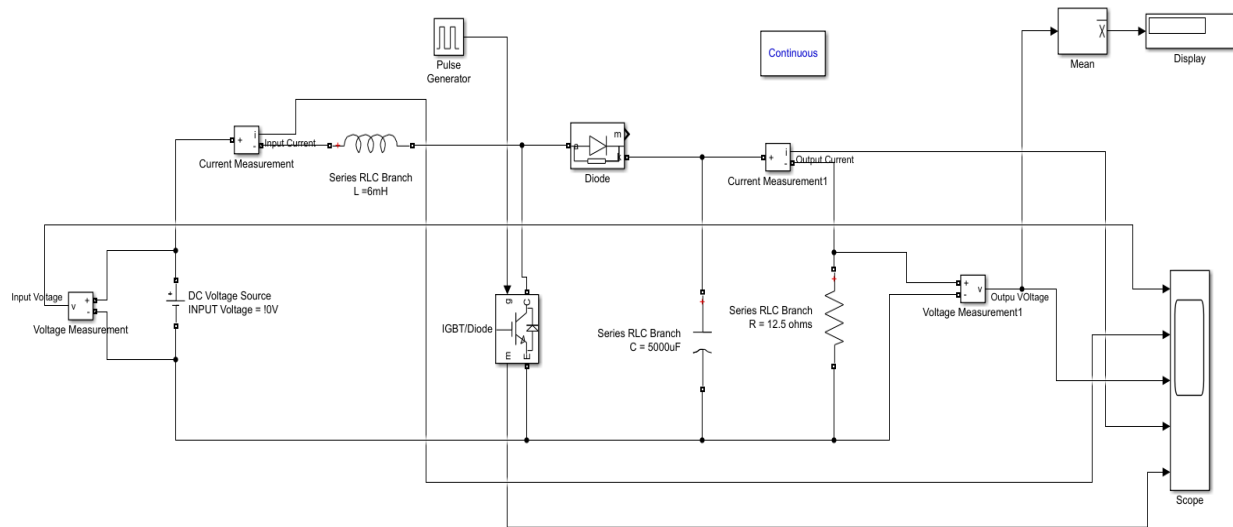
## I Block Diagram of Buck Converter:



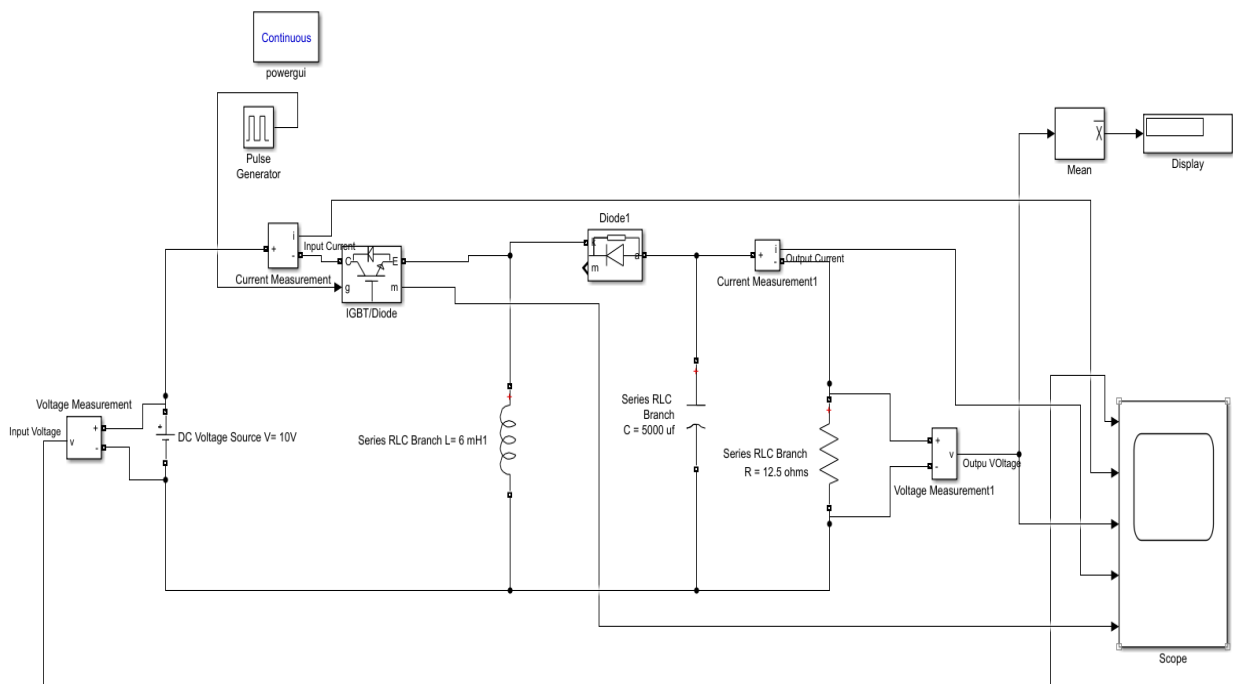
## Pulse generator showing 40% duty cycle



## II Block Diagram of Boost Converter:



## III Block Diagram of Buck-Boost Converter:



**Conclusion:** Obtain the results as per “Exp8\_Part B.doc” file.